be zero and the later time to be *t*. The state at time *t* is given by some operation that we call **U**(*t*), acting on the state at time zero. Without further specifying the properties of **U**(*t*), this tells us very little except that |Ψ(*t*)> is determined by |Ψ(0)>. Let's express this relation with the equation

$$|\Psi(t)\rangle = \mathbf{U}(t)|\Psi(0)\rangle. \qquad (4.1)$$

The operation **U** is called *the time-development operator* for the system.

4.3 DETERMINISM IN QUANTUM MECHANICS

At this point, we need to draw some careful distinctions. We are setting up **U**(*t*) in such a way that the state-vector will evolve in a deterministic manner. Yes, you heard me correctly—the time evolution of the state-vector is *deterministic*. This is nice because it provides us with something we can try to predict. But how does that square with the statistical character of our measurement results?

As we've seen, knowing the quantum state does not mean that you can predict the result of an experiment with certainty. For example, knowing that the state of a spin is |*r*> may tell you the outcome of a measurement of σ_x but tells you nothing about a measurement of σ_z or σ_y. For this reason, Eq. 4.1 is not the same as classical determinism. Classical determinism allows us to predict the results of experiments. The quantum evolution of states allows us to compute the probabilities of the outcomes of *later* experiments.

This is one of the core differences between classical and quantum mechanics. It goes back to the relationship between states and measurements we mentioned at the very beginning of this book. In classical mechanics, there's no real difference between states and measurements. In quantum mechanics, the difference is profound.

4.4 A CLOSER LOOK AT U(*T*)

Conventional quantum mechanics places a couple of requirements on **U**(*t*). First, it requires **U**(*t*) to be a linear operator. That is not very surprising. The